

Phase Equilibria in the ZnO-“FeO”-SiO₂ System in Reducing Atmosphere and in the ZnO-“FeO”-SiO₂-“Cu₂O” System in Equilibrium with Liquid Copper Metal at 1250 °C (1523 K)



TAUFIQ HIDAYAT, PETER C. HAYES, and EVGUENI JAK

Recent experimental studies in the ZnO-“FeO”-SiO₂ system in reducing atmosphere demonstrated significant discrepancies with the current FactSage thermodynamic model developed using previous experimental data in this system in equilibrium with metallic iron and air. The present experimental study on phase equilibria in the ZnO-“FeO”-SiO₂-“Cu₂O” system in equilibrium with liquid copper at 1250 °C (1523 K) at low copper oxide concentrations in slag was initiated and undertaken to resolve these discrepancies. A high-temperature equilibration-rapid quenching-electron-probe X-ray microanalysis (EPMA) technique using a primary phase substrate support and closed system approach with Cu metal introduced to determine effective equilibrium oxygen partial pressure from the Cu_{metal}/Cu₂O_{slag} equilibria was applied to provide accurate information on the liquidus and corresponding solid compositions in the spinel, willemite, and tridymite primary phase fields. The present results confirmed the accuracy of the FactSage model, resolved discrepancies, and demonstrated significant uncertainties in the recent studies by other authors on the system in the open reducing atmosphere. The present study shows how this closed system approach can be used to obtain key thermodynamic data on phase equilibria in systems containing volatile metal species, overcoming the limitations and uncertainties encountered in conventional open gas/condensed phase equilibration with these systems. The study highlights the importance of the focus on obtaining accurate experimental data and the risks of misleading information from inadequate experimental control and analysis. The study also demonstrates that continuing in-depth critical review and analysis of the elemental reactions taking place in complex systems is an essential step in phase equilibrium research.

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I. INTRODUCTION

ZINC is present in lead, zinc, and many copper industrial smelting processes operated over a wide range of process conditions, from oxidizing to highly reducing conditions, and from low to high temperatures; these are complex chemical systems. Further improvement and optimization of the industrial processes requires support in the form of detailed thermodynamic and thermochemical information. These data can be presented in many ways; optimized multicomponent multiphase thermodynamic databases in conjunction with

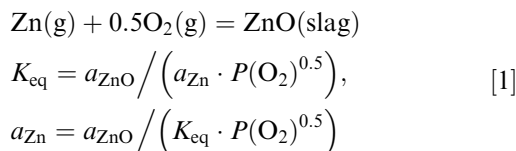
predictive computer programs, such as FactSage (developed by CRCT (Montreal) and GTT-Technologies (Herzogenrath, Germany)),^[1] are increasingly used in research and industry. The databases are, however, only as good as the quality of experimental data that are used in their construction; accurate experimental data are absolutely essential if accurate thermodynamic databases are to be developed, particularly essential are accurate data on low-order binary and ternary subsystems that form the foundation of the multicomponent databases. Adopting inaccurate experimental data is of critical concern since this can lead to distortions in the databases, misleading model predictions, and incorrect technological/operational decisions in industrial practice. All data used in the database development, therefore, are critically reviewed, particularly in relation to the suitability of the experimental methodology used, measurement techniques, and analysis before incorporation into the database.

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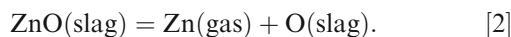
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A. Previous Studies on Zn-Containing Slag Systems in Equilibrium with Metallic Fe and Air Based on Microanalysis

The principal zinc-containing species in the gas phase are Zn (g) and ZnO (g).^[2] The saturation equilibrium vapor pressure of Zn metal, $P^\circ(\text{Zn})$, is high and the ZnO (g) partial pressure is very low.^[2] The zinc partial pressure, $P(\text{Zn (g)})$, therefore, is high at high temperatures and low oxygen partial pressures and increases significantly with decreasing $P(\text{O}_2)$ for a given ZnO concentration and activity in slag according to the reaction



Note that $a_{\text{Zn}} = P(\text{Zn})/P^\circ(\text{Zn})$. The vaporization of Zn (g) from the ZnO-containing slag system in reducing conditions, therefore, rapidly takes place predominantly as Zn (g), thus leaving remaining oxygen in the slag phase according to the simplified reaction



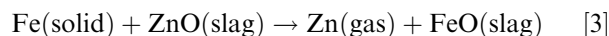
It is well established from the results of previous experimental studies on activity measurements and phase equilibria^[3–7] in zinc-containing slag systems that achieving true chemical equilibrium between flowing gas and zinc-containing liquid slags is highly problematic and practically impossible for high-temperature and low oxygen potential conditions. While the gas mixtures, such as CO/CO₂, can be used to control the effective oxygen pressure in the gas phase, similar levels of control over the zinc partial pressure in the condensed slag and solid phases are difficult if not impossible to achieve in practice. If there is no zinc in the flowing gas stream, then there will always be a net driving force for zinc vaporization and, therefore, zinc will be continuously removed from the slag as Zn (g), leaving behind O in slag. The effective oxygen partial pressure in the condensed system under these conditions remains uncertain. All phase equilibrium studies using an open system flowing gas approach are subject to these uncertainties.

An experimental approach to resolve uncertainties caused by zinc vaporization involving high-temperature equilibration followed by the rapid quenching technique and microanalysis of phases was developed by Jak *et al.*^[8–10] to measure phase equilibria in the Zn-containing slag systems in equilibrium with metallic Fe, including the ZnO–“FeO”–SiO₂ system. The compositions of the phases after equilibration were measured using electron-probe X-ray microanalysis (EPMA). High-ZnO master slags were prepared first in air. The master slags were then equilibrated in contact with excess of metallic Fe powder. In this way, the oxygen partial pressure was not controlled by the gas-slag equilibration, but instead by maintaining the activity of iron at unity relative to the pure-solid standard state. Therefore, the condensed phases-equilibration rather

than gas-equilibration approach was used for this ZnO-containing system.

That experimental approach followed analysis of the number of degrees of freedom of the system expressed as $F = C + 2 - P$, where C = number of component and P = number of phases. In the ZnO–“FeO”–SiO₂ system, $F = 4$ (Zn, Fe, O, Si) + 2 – 4 (gas, liquid slag, solid oxide, metallic Fe) = 2. The number of degrees of freedom $F = 0$ for the fixed temperature and ZnO concentration in slag (the latter was measured by EPMA after the experiment) so that the system was fully specified.

During slag/metal equilibration, the following reaction takes place:



resulting in the formation of the ZnO–“FeO”–SiO₂ slag from the initial high-zinc-silicate master slag and Fe powder. The equilibrium between the condensed phases and metallic Fe was readily achieved, unlike the equilibrium between gas and condensed phases. Particular attention was given to the confirmation of achievement of equilibrium in those studies^[8–10] using advanced microanalysis capabilities, including systematic phase compositions measurements as a function of distance from the phase interfaces at microscale as well as compositional profiles across entire samples at macroscale.

Those studies^[8–10] were the first systematic characterizations of liquidus in the ZnO-containing system at reducing conditions. The experimental data on phase equilibria in the ZnO-containing systems at reducing conditions along with the systematic experimental data in air^[11] were then used to develop the FactSage thermodynamic database for the PbO–ZnO–SiO₂–CaO–FeO–Fe₂O₃ system,^[11–15] which is part of the current FactSage public thermodynamic database.^[16]

B. Recent Phase Equilibria Studies in the Zn-Containing Slag Systems at Intermediate Oxygen Partial Pressures

Liu *et al.*^[17–21] recently reported phase equilibria results of experiments for the ZnO–“FeO”–SiO₂ and ZnO–“FeO”–SiO₂–Al₂O₃ systems performed at $P(\text{O}_2) = 10^{-8}$ atm controlled by flowing CO/CO₂/Ar gas mixtures, at temperatures between 1170 °C to 1300 °C (1443 K to 1573 K). Zinc silicate master slag prepared in air at 1500 °C (1773 K) and pre-equilibrated iron silicate master slag were used as starting materials. Liu *et al.* argued that equilibrium was achieved within short time (tens of minutes) since the pre-equilibrated zinc-silicate and iron-silicate slags were used in the starting mixture. The only evidence given by Liu *et al.* to support the claim of achievement of equilibrium was that the zinc concentration in the liquid slag was uniform in composition. This is a necessary but not sufficient condition to prove that the gas/slag equilibrium was achieved. It only indicates that the mass transfer of zinc in the slag is rapid relative to other reaction steps.

The data reported by Liu *et al.*,^[17–21] and shown in Figure 1, indicate that the spinel liquidus in the ZnO–“FeO”–SiO₂ system at $P(\text{O}_2) = 10^{-8}$ atm moves toward higher silica concentrations with increasing ZnO concentration in slag, meeting directly with the tridymite (SiO₂) liquidus at approximately 0.35SiO₂/(ZnO + “FeO” + SiO₂) by weight. These results by the other authors are in marked contrast with predictions for these conditions using the current FactSage thermodynamic public database^[11–16] developed by the present author, and with previous experimental results obtained for slags in equilibrium with metallic iron^[8–10] and in equilibrium with air^[11] obtained by the present authors. The behavior predicted using the optimized thermodynamic database also indicates that the relative stability of the spinel phase is strongly influenced by the oxygen partial pressure of the system; the spinel phase becomes more stable relative to the liquid phase with increasing oxygen partial pressure.

Present experimental investigation of the ZnO–“FeO”–SiO₂ system at intermediate oxygen partial pressures was, therefore, initiated to resolve the differences between the recent results by Liu *et al.*^[17–21] and FactSage predictions.^[11–16] It was essential to do this study to ensure accurate information is used in multi-component thermodynamic database development for FactSage users. In the present study, an alternative methodology for accurately measuring condensed phase equilibria was used for the system having high vapor pressure of Zn metal species.

II. PRESENT EXPERIMENTAL METHODOLOGY

A. General Description of Experimental Methodology

The experimental methodology adopted by the co-authors^[22,23] in the present study involves high-temperature equilibration at controlled conditions, rapid

quenching, and direct measurement of the compositions of equilibrium phases with EPMA.

The samples in the present study were equilibrated on prefabricated substrates made of the primary solid phases. The samples were suspended on a wire and equilibrated at 1250 °C (1523 K) in a vertical impervious alumina ceramic tube in an electrically heated resistance furnace in controlled atmospheres. The furnace temperature was monitored by an alumina-shielded Pt/Pt-Rh 13 pct thermocouple placed immediately adjacent to the sample. The thermocouple was periodically calibrated against a standard thermocouple. After the designated equilibration time, the samples were quenched into cold water and mounted in resin; then, polished cross sections were prepared and carbon coated using a QT150TES carbon coater (Quorum Technologies, United Kingdom) for further analysis.

The compositions of the individual phases were measured using an electron-probe X-ray microanalyzer, JEOL* JXA 8200L. An acceleration voltage of 15 keV

* JEOL is a trademark of Japan Electron Optics Ltd., Tokyo.

and a probe current of 15 nA were used. The Duncumb–Philibert ZAF correction procedure supplied with the JEOL JXA 8200L probe was applied. Pure oxide and metal standards from Charles M. Taylor (Stanford, CA) and purchased oxide standards from Structure Probe, Inc. (PA), which include zincite (ZnO), hematite (Fe₂O₃), quartz (SiO₂), cuprite (Cu₂O), Cu metal, and Fe metal, were used. Only the concentrations of the metal cations in the phases were measured with EPMA; the phase compositions were recalculated to oxides using selected oxidation states for presentation purposes and to provide unambiguous reporting.

A four-points-test approach introduced by one of the authors^[23] was applied to ensure achievement of equilibrium and includes the following: (1) variation of equilibration time, (2) analysis of phase homogeneity, (3) approaching final equilibrium points from different directions, and (4) systematic analysis of reactions specific to the investigated system. Confirmation of equilibrium is important, especially for studies of such intricate ZnO-containing systems at reducing conditions. The systematic analysis of reactions specific to the system is essential to identify possible uncertainties and develop adequate techniques suitable to the particular investigated system. Based on this systematic analysis, special modifications and developments of specific methodology suitable to the particular system are introduced and implemented.

B. Analysis of Reactions Taking Place during Equilibration in the Present System

Figure 2(a) illustrates the reactions taking place during equilibration of the ZnO–FeO–Fe₂O₃–SiO₂ slag with gas. The targeted gas-slag reaction in the open gas equilibration experiment with the Fe-oxide containing slag equilibrated under flowing CO/CO₂ gas atmosphere is

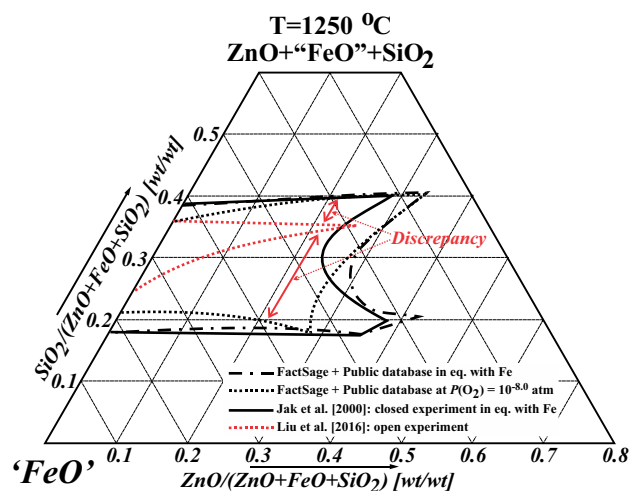
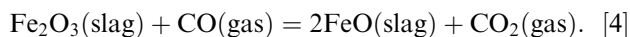
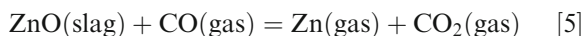
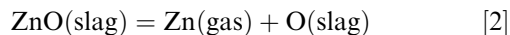


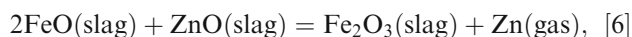
Fig. 1—Liquidus isotherms at 1250 °C (1523 K) for the ZnO–“FeO”–SiO₂ system in equilibrium with metallic Fe and at $P(\text{O}_2) = 10^{-8}$ atm. Calculated lines from FactSage using the public database.^[11–16] Experimental data from Jak *et al.*^[8] and Liu *et al.*^[19]



The ferric to ferrous ratio in the slag determines the effective oxygen partial pressure in the condensed system. In the ZnO-containing slag in reducing conditions, the open system equilibration approach leads to continuing removal of zinc from the condensed phases through vaporization of the Zn gaseous specie that can be described by the following reactions:

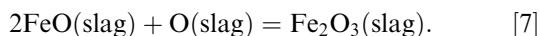


and



where Reaction [5] may be viewed as an effective buffer at the slag surface restricting the bulk slag from the reaction with the CO/CO₂ gas, and Reaction [2] may be viewed as an effective “pump” of the oxygen left by ZnO during Zn (gas) vaporization into the bulk slag.

Reaction [6], therefore, may be viewed as a combination of Reactions [2] and [7]:



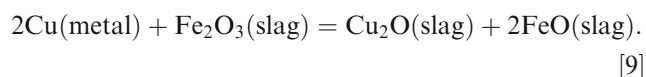
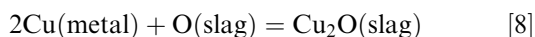
These processes may be expressed simply by the following statement: As zinc vaporizes from the slag, the effective oxygen partial pressure of the slag phase increases. The higher the ZnO concentration in the initial mixture, the more significant is the effect.

C. Adoption of Experimental Methodology for the ZnO-Containing Slag System

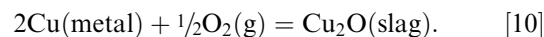
1. $\text{Cu}_{\text{metal}}/\text{Cu}_2\text{O}_{\text{slag}}$ couple as internal measure of oxygen partial pressure

An experimental methodology suitable for the ZnO-“FeO”-SiO₂ system at intermediate oxygen partial pressures was developed. To overcome uncertainties associated with the Zn vaporization, the experimental methodology involves the use of a sealed system with the $\text{Cu}_{\text{metal}}/\text{Cu}_2\text{O}_{\text{slag}}$ couple used as internal measure of the effective oxygen partial pressure.

A schematic illustrating the important elementary reactions taking place in the ZnO-FeO-Fe₂O₃-SiO₂-“Cu₂O” system in equilibrium with metallic Cu is provided in Figure 2(b). In the presence of metallic Cu, a change in the oxidation state of the slag results in a change in the “Cu₂O” solubility in liquid slag through the following reactions:



The $\text{Cu}_{\text{metal}}/\text{Cu}_2\text{O}_{\text{slag}}$ interaction reflects the effective oxygen partial pressure of the slag phase, as indicated by the following reaction:



Based on this reaction, it is possible to derive the effective $P(\text{O}_2)$ in the slag from the concentration of “Cu₂O” in the liquid slag phase measured with EPMA. This, however, requires calibration data of “Cu₂O” in slag vs $P(\text{O}_2)$ for the ZnO-“FeO”-SiO₂-“Cu₂O” system. In the present study, the following approach was developed and applied, involving the following.

1. Establishing a calibration relationship for the “Cu₂O” chemically dissolved in slag vs $P(\text{O}_2)$ in the “FeO”-SiO₂-“Cu₂O” system using accurate experimental data obtained with the open gas-equilibration technique (this relationship can be accurately measured since the system contains no high-vapor pressure metal species).
2. Extrapolating the “Cu₂O” in slag vs $P(\text{O}_2)$ calibration data to the ZnO-containing system based on the trend predicted by FactSage with the PYROSEARCH thermodynamic database^[24] based on the latest experimental data.
3. Performing equilibration of the slag in the ZnO-“FeO”-SiO₂-“Cu₂O” system with the Cu metal on a substrate in a sealed silica ampoule and measurement of the “Cu₂O” in slag in equilibrium with the Cu metal using EPMA.
4. Deriving the effective internal $P(\text{O}_2)$ in the condensed phases in the sample from the “Cu₂O” in slag measured with EPMA (step 3) and the information obtained in the preceding steps 1 and 2.

The variation of the activity coefficient of “Cu₂O” in Eq. [10] due to the variation of slag composition was taken into account in steps 1 and 2 of this approach. At these low concentrations of copper in slag (between 3 and 5 wt pct “Cu₂O”), the activity coefficient of “Cu₂O” in the slag taken from the database^[24] is in the range 0.50 to 1.15 depending on the slag composition.

A special series of experiments in the “FeO”-SiO₂-“Cu₂O” system with controlled gas atmosphere was undertaken in the present study to accurately establish “Cu₂O” vs $P(\text{O}_2)$. The experimental details will be described subsequently. The calibration data of “Cu₂O” in slag vs $P(\text{O}_2)$ for the “FeO”-SiO₂-“Cu₂O” system at 1250 °C at slag/copper/tridymite and slag/copper/spinel equilibria in the present study together with two experimental points from Henao *et al.*^[25] are given in Figures 3(a) and (b), respectively. The experimental data are compared with the calculated values using FactSage with the current thermodynamic database,^[24] demonstrating agreement with the experimental “Cu₂O” in slag vs $P(\text{O}_2)$ for the slag/copper/tridymite equilibrium and within ~ 0.5 wt pct for the slag/copper/spinel equilibrium. The uncertainty of oxygen partial pressure obtained from the present approach is estimated to be within ~ 0.05 log₁₀[$P(\text{O}_2)$, atm] unit.

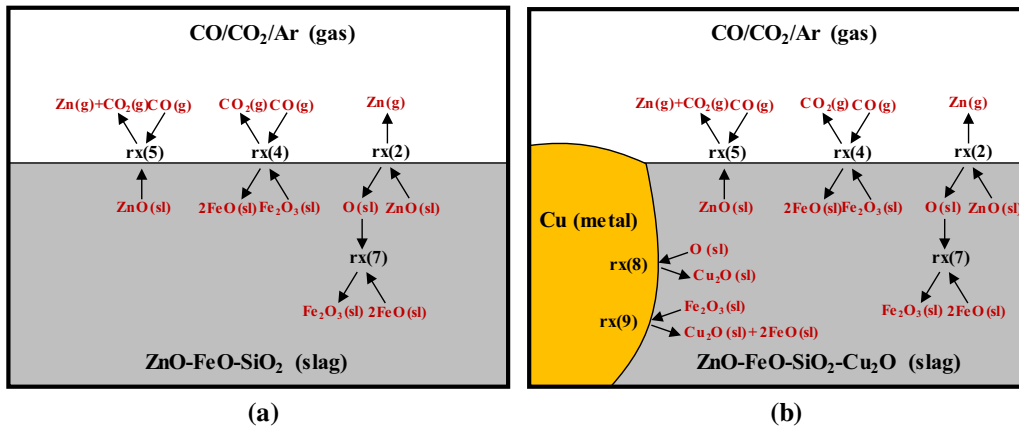


Fig. 2—Schematic of the important elementary reactions taking place in the (a) ZnO-FeO-Fe₂O₃-SiO₂ and (b) ZnO-FeO-Fe₂O₃-SiO₂-“Cu₂O” system in equilibrium with metallic Cu.

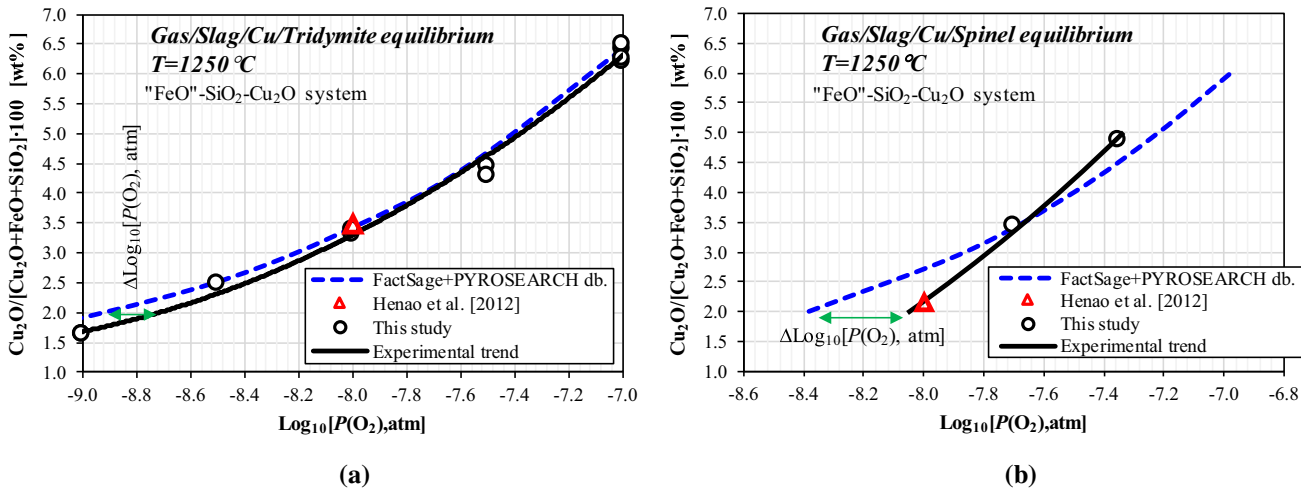


Fig. 3—Calibration data of “Cu₂O” in slag vs $P(\text{O}_2)$ in the “FeO”-SiO₂-“Cu₂O” system at 1250 °C (1523 K): (a) at slag/copper/tridymite equilibrium and (b) at slag/copper/spinel equilibrium. Dashed blue lines are from FactSage predictions with the PYROSEARCH database.^[24] Points are experimental data from the present study and Henao *et al.*^[25] Bold black lines are experimental trends.

The “Cu₂O” vs $P(\text{O}_2)$ trends given in Figure 3 were used only to establish the relationship between the “Cu₂O” concentration and $P(\text{O}_2)$ in the ZnO-free slag (step 1 of the approach), and the effect of ZnO addition to the “FeO”-SiO₂ slag on the $P(\text{O}_2)$ vs “Cu₂O” relationship was taken into account by FactSage predictions (step 2 of the approach).

2. Initial starting materials

The iron-silicate master slag and synthesized willemite compound were used in the starting mixture to enhance the rate of attainment of equilibrium. The iron-silicate master slag was prepared by mixing high-purity Fe, Fe₂O₃, and SiO₂ powders (99.9 wt pct supplied by Alpha Aesar) and equilibration in platinum foil at 1250 °C (1523 K) for 18 hours in the CO/CO₂/Ar gas stream corresponding to $P(\text{O}_2) = 10^{-8}$ atm. The willemite (Zn₂SiO₄) was prepared by equilibrating a stoichiometric ZnO-SiO₂ powder mixture at 1450 °C (1723 K) in air for 3 hours. The compositions and microstructures of the iron-silicate master slag and partially sintered mixture corresponding to the willemite are shown in

Table I and Figure 4, respectively. The figure shows that the master slag and the synthesized willemite contain small proportions of other solid phases, spinel and zincite, respectively.

The iron-silicate master slag and synthetic willemite solid were crushed to powders of approximately less than 10-μm particle size in an agate mortar and pestle. An excess of copper metal powder was added to the mixture to ensure the presence of copper metal in the final samples and to obtain equilibrium between the liquid slag and liquid copper, resulting in the dissolution of “Cu₂O” into the slag. The “Cu₂O” concentration dissolved in the slag measured with EPMA after experiment was subsequently used as an internal measure of the effective $P(\text{O}_2)$ of the slag. The higher the “Cu₂O” concentration in the slag, the higher the effective $P(\text{O}_2)$ of the slag, as indicated by Reaction [10]. The “Cu₂O” concentration and the effective $P(\text{O}_2)$ of the slag within the sealed ampoule were controlled by adjusting the compositions of the initial mixtures, either by the addition of “Cu₂O” in the initial mixture to increase the resulting $P(\text{O}_2)$ or by the addition of Fe

Table I. Initial Mixtures Used in the Gas/Slag/Copper/Solid Equilibrium Experiment

Exp. Code	Temperature [°C (K)]	Time (h)	Condition	Normalized Oxide Composition (Wt Pct)			Old Total
				ZnO	“FeO”	SiO ₂	
Master slag	1250 (1523)	18	$\log_{10}[P(\text{O}_2), \text{atm}] = -8$	0.08	74.2	25.7	99.1
Zn ₂ SiO ₄	1450 (1723)	3	in air	71.9	0.00	28.1	101.9

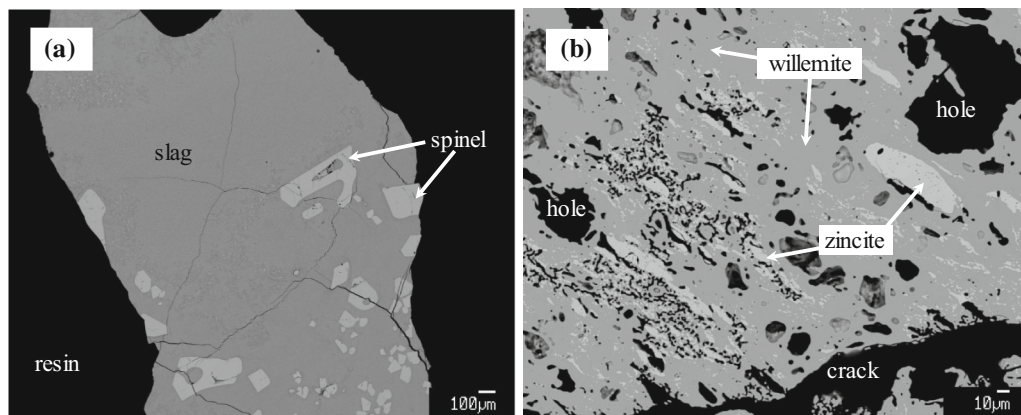


Fig. 4—SEM backscattered image of starting materials: (a) master slag from equilibration at 1250 °C (1523 K) and $\log_{10}[P(\text{O}_2), \text{atm}] = -8$ for 18 h and (b) partially sintered Zn₂SiO₄ from annealing at 1450 °C (1723 K) for 3 h.

metal in the initial mixture to decrease the resulting $P(\text{O}_2)$. Small proportions of pure Fe, FeO, ZnO, SiO₂, and “Cu₂O” powders, therefore, were added to adjust the initial composition to ensure that (a) several phases and (b) predetermined “Cu₂O” concentration in the liquid slag were obtained in the samples after equilibration.

3. Closed system equilibration

The initial mixture was pelletized and placed on a substrate made of the primary phase solid. The sample was sealed at room temperature in an evacuated silica tube (99.9 wt pct supplied by Lianyungang Guoyi Quartz Products Co. Ltd., China) using an oxy-hydrogen gas burner to soften the silica and rotary vacuum pump to remove the gas from the tube. In this closed system equilibration experiment, insignificant zinc loss from the condensed phases to the gas takes place due to the limited ampoule volume. In the case of equilibrium with tridymite, the initial mixture was placed directly in a sealed silica ampoule. In the case of equilibrium with spinel, the mixture was placed on a specially prepared spinel substrate (iron foil oxidized at fixed condition) and fitted inside the sealed silica ampoule. Contact between spinel substrate and the inner wall of silica ampoule was restricted by the use of platinum foil. The ampoule was suspended in the uniform hot zone of a furnace during equilibration. Following equilibration, the sample was cooled rapidly by quenching into a bucket filled with iced brine. Metal block was placed at the base of the bucket to break

ampoules in order to increase the quenching rate. The quality of quenching and the lack of corresponding uncertainties were confirmed in samples by control of phase homogeneity and measuring glass (quenched liquid) compositional profiles as a function of distance from solid phases.

D. Open System vs Closed System Preliminary Equilibrations to Investigate Elemental Processes during Experiments

Preliminary experiments were carried out to investigate the slag/copper/spinel equilibration process in the ZnO-“FeO”-SiO₂-“Cu₂O” system at 1250 °C (1523 K), $P(\text{O}_2) = 10^{-8}$ atm through the following:

- open system equilibration (slag sample on the substrate directly exposed to the CO/CO₂/Ar gas with total flow rate of 500 mL/min continuously flushed through the reaction tube with 32-mm inner diameter); and
- closed system equilibration (slag sample on the substrate is held in the preliminary evacuated and then hermetically sealed silica ampoule).

A micrograph of the sample cross section obtained from the open experiment is given in Figure 5; the compositions of the slag in the sample measured at different locations—at the slag-Cu interface (A slag), at the gas-slag interface far from Cu (B slag), and within the bulk slag far from Cu (C slag)—are given in Table II.

The result obtained from the open system equilibration shows the following.

1. Significant ZnO removal was observed in 0.5 hours—from 38 wt pct ZnO in the starting mixture to 5 wt pct ZnO in the final slag. It was not possible to maintain considerable Zn concentration in the slag for equilibration time longer than 0.5 hour; this was mainly due to the Zn vaporization (ZnO concentration exponentially decreased with increasing equilibration time) and slightly due to ZnO dissolution into the spinel substrate. The closed system approach restricted the Zn vaporization and enabled the equilibration to be carried out for longer time. These results indicate that in the previous studies by Liu *et al.*,^[17–21] the starting compositions were not reported but seemed to include higher ZnO in the initial slag or shorter equilibration time to obtain higher ZnO in the final samples.
2. There was no detectable concentration gradient of ZnO in the slag, indicating relatively fast mass transfer of ZnO from the bulk to the surface of the slag, which promotes Zn vaporization into the gas phase.
3. There were higher “Cu₂O” concentrations close to the slag–Cu metal interface, indicating the following:

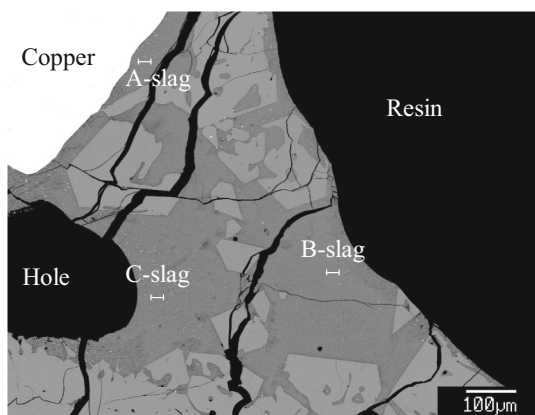


Fig. 5—SEM backscattered image showing the microstructure of the sample from the open system gas/slag/copper/spinel equilibration (0.5 h) in the ZnO-“FeO”-SiO₂-“Cu₂O” system at 1250 °C (1523 K) and the location of the measurements at the slag–Cu interface (A), at the gas–slag interface far from Cu (B), and within the bulk slag far from Cu (C).

(1) the Reaction [10] Cu (metal) + O (slag) = Cu₂O (slag) taking place, (2) relatively fast oxygen transfer from the surface into bulk slag, and (3) relatively slow “Cu₂O” mass transfer through the liquid slag phase.

4. From the information on the “Cu₂O” concentration in the slag measured by EPMA, it was shown that the slag had higher effective $P(O_2)$ compared to that of the CO/CO₂ in the gas phase. This strongly supports the argument that the presence of excess oxygen is created by the Zn vaporization, resulting in the uncertainties for the open system equilibration of the ZnO-containing system.

In contrast, the closed system experiment demonstrates that longer equilibration time is possible and uniform “Cu₂O” concentration across the entire sample is obtained.

The experimental result shows that although the liquid and solid compositions can be accurately measured using EPMA, the achievement of equilibrium at controlled conditions requires significant further efforts; for example, the oxygen partial pressures present in this system are uncertain in the case of the open system–gas–equilibration approach. Introduction of Cu metal and the closed equilibration approach to control Cu metal activity, rather than oxygen potential, enable accurate characterization of this complex system. For this reason, all of the subsequent data reported for the slag/copper/solid equilibrium in the ZnO-“FeO”-SiO₂-“Cu₂O” system in the present study were obtained using the closed equilibration technique and the Cu_{metal}/Cu₂O_{slag} “internal standard” with equilibration times longer than 2 hours.

III. RESULTS

SEM micrographs of typical quenched samples from the closed slag/copper/solid equilibration experiments in the ZnO-“FeO”-SiO₂-“Cu₂O” system in equilibrium with metallic Cu at 1250 °C (1523 K) are given in Figure 6. Figures 6(a) and (b) show liquid slag in equilibrium with liquid Cu and one primary solid phase, *i.e.*, tridymite and spinel, respectively. Figures 6(c) and (d) show liquid slag in equilibrium with liquid Cu and two solid phases, *i.e.*, tridymite + willemite and spinel + willemite, respectively.

Table II. Measured Compositions of Liquid Slag from Gas/Slag/Copper/Solid Equilibrium in the ZnO-“FeO”-SiO₂-“Cu₂O” System at 1250 °C (1523 K) from the Open System Experiment at $P(O_2) = 10^{-8}$ Atm

Technique of Equilibration	Time (h)	Location	Normalized Oxide Composition (Wt Pct)				Old Total	Estimated Log ₁₀ [$P(O)$, Atm]*
			ZnO	“FeO”	SiO ₂	“Cu ₂ O”		
Open system	0.5	A slag	5.0	69.3	22.4	3.2	99.3	– 7.67
		B slag	5.1	69.4	22.4	3.0	99.3	—
		C slag	4.9	69.6	22.8	2.8	99.5	—

*Estimated $P(O_2)$ based on calibration curve in Fig. 3 and FactSage prediction using PYROSEARCH database^[24] (Table III).

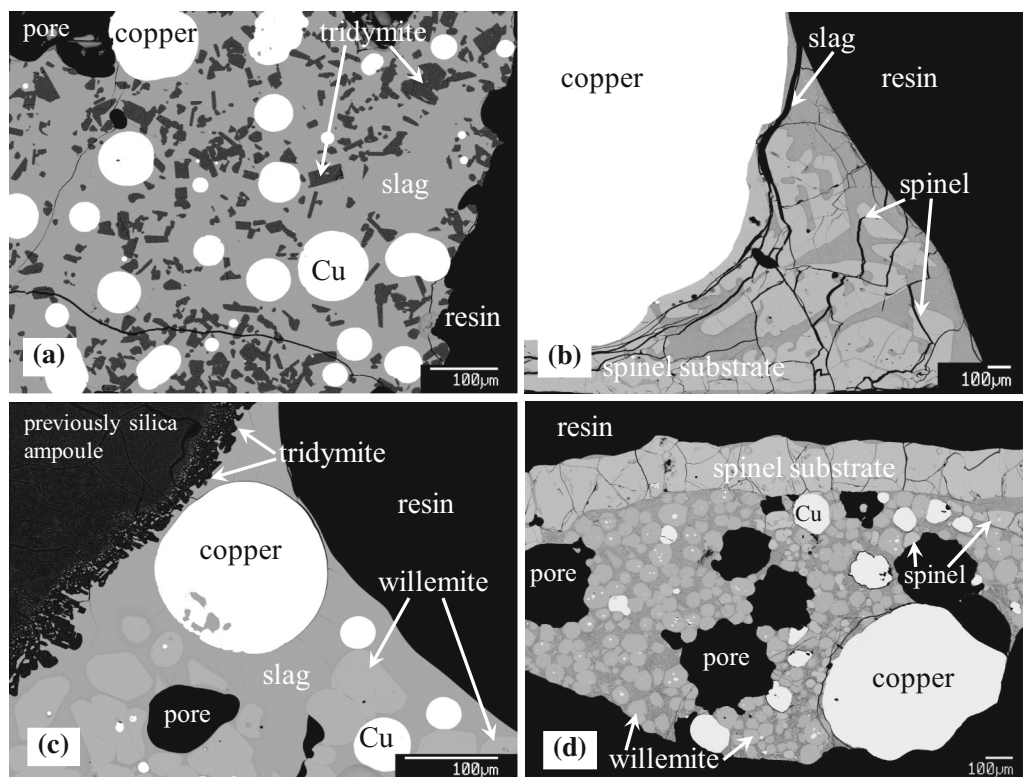


Fig. 6—SEM backscattered electron image of samples from gas/slag/copper/solid equilibria in the ZnO-“FeO”-SiO₂-“Cu₂O” system at 1250 °C (1523 K): (a) liquid slag + tridymite (SiO₂), (b) liquid slag + spinel (Fe₃O₄), (c) liquid slag + tridymite (SiO₂) + willemite (Zn₂SiO₄), and (d) liquid slag + spinel (Fe₃O₄) + willemite (Zn₂SiO₄).

Table III shows examples of the estimation of effective $P(\text{O}_2)$ for the ZnO-containing system based on the approach used in the present study. First, $\log_{10}[P(\text{O}_2), \text{atm}]$ for the ZnO-containing system was calculated using FactSage with PYROSEARCH database^[24] for the experimental points given in rows *a* and *k* in Table III (steps *b* and *l*). Then, $\log_{10}[P(\text{O}_2), \text{atm}]$ for the ZnO-free system for the same “Cu₂O” in slag was calculated using the calibration curves shown in Figure 3 (Table III, steps *c* and *m*) and FactSage with PYROSEARCH database^[24] (Table III, steps *d* and *n*). Finally, $\log_{10}[P(\text{O}_2), \text{atm}]$ of the ZnO-containing system from FactSage prediction was calibrated/corrected based on the difference between $\log_{10}[P(\text{O}_2), \text{atm}]$ of the ZnO-free system obtained from the calibration curve and FactSage prediction (Table III, *e* and *o*).

Table IV provides the results of the experiments—the compositions of liquid slag, solid oxide, and liquid metal phases measured with EPMA. Experiments 1 through 13 are results from the special experiments in the “FeO”-SiO₂-“Cu₂O” system with controlled gas atmosphere (open equilibration) undertaken to accurately establish “Cu₂O” vs $P(\text{O}_2)$ for the ZnO-free system. Experiments 14 through 23 in Table IV are experimental results for the ZnO-containing system. The effective $P(\text{O}_2)$ ’s of the ZnO-containing system, estimated using the preceding approach, are also given in Table IV. Experiments 18 and 19 in Table IV show that experiments with high ZnO concentration in slag can be done

for equilibration time of 3 hours and even as long as 18 hours without significant ZnO loss from the slag. These two experiments gave the same tridymite liquidus position (Figure 9), indicating that the equilibration process of samples in the closed system can be done in relatively short equilibration time.

Figure 7 shows “Cu₂O” concentration in slag as a function of $\log_{10}[P(\text{O}_2), \text{atm}]$ for the tridymite and spinel primary phase fields at fixed $[\text{Zn} + \text{Fe}]/\text{SiO}_2$ ratios and different Zn concentrations in slag based on FactSage predictions after the calibration process described in Table III. In general, the effective $\log_{10}[P(\text{O}_2), \text{atm}]$ increases with increasing Zn concentration in the slag in both primary phase fields.

The complex ZnO-FeO-Fe₂O₃-SiO₂-“Cu₂O” slag system in equilibrium with metallic Cu can be represented in a simplified ZnO-“FeO”-SiO₂ pseudo-ternary diagram at fixed “Cu₂O” in slag. The projection of the slag system onto the ZnO-“FeO”-SiO₂ plane through the O corner is illustrated in Figure 8. This is done by recalculating all iron in the slag as FeO and normalizing the slag composition to the ZnO-“FeO”-SiO₂ ternary.

Figure 9 shows the projected liquid and solid compositions in equilibrium with liquid copper at 1250 °C (1523 K) and fixed “Cu₂O” concentrations in slag. The label adjacent to each data point shows the measured “Cu₂O” concentration in the slag. Red lines and blue lines are the projected liquidus and equilibrium solid compositions at 3.4 wt pct “Cu₂O” ($P(\text{O}_2) = \sim 10^{-7.6}$

Table IV. EPMA Measured Compositions of Phases from Gas/Slag/Copper/Solid Equilibration in Open and Closed Systems for the ZnO-“FeO”-SiO₂-“Cu₂O” System at 1250 °C (1523 K)

Exp. Code	Time (h)	Phase	Normalized Oxide Composition (Wt Pct)				Old Total	(Zn + Fe)/ SiO ₂ in Slag	Phase	Normalized Metal Composition (Wt Pct)			Old Total	Log ₁₀ [<i>P</i> (O ₂), Atm]
			ZnO	"FeO"	SiO ₂	"Cu ₂ O"				Zn	Fe	Cu		
ZnO-free system														
1	18 [†]	slag	0.0	67.9	27.2	4.9	98.9	1.94	copper	0.00	0.00	99.9	100.2	− 7.35**
		spinel	0.0	99.2	0.63	0.15	92.4							
2	24 [†]	slag	0.0	71.2	25.3	3.5	98.8	2.19	copper	0.00	0.03	99.9	100.7	− 7.70**
		spinel	0.0	99.3	0.61	0.09	93.0							
3	24 [†]	slag	0.0	61.1	32.5	6.4	100.3	1.46	copper	0.0	0.02	99.9	99.4	− 7.00**
		tridymite	0.0	0.55	99.4	0.05	100.3							
4	24 [†]	slag	0.0	61.2	32.3	6.5	100.2	1.47	copper	0.0	0.02	99.9	101.4	− 7.00**
		tridymite	0.0	0.48	99.5	0.06	99.7							
5	48 [†]	slag	0.0	60.8	33.0	6.2	101.1	1.43	copper	0.0	0.01	99.9	101.2	− 7.00**
		tridymite	0.0	0.59	99.4	0.04	101.7							
6	24 [†]	slag	0.0	60.8	32.9	6.2	100.5	1.44	copper	0.0	0.02	99.9	99.1	− 7.00**
		tridymite	0.0	0.42	99.5	0.10	100.8							
7	24 [†]	slag	0.0	61.1	34.5	4.5	99.1	1.38	copper	0.0	0.03	99.9	99.0	− 7.50**
		tridymite	0.0	0.69	99.2	0.08	99.8							
8	18 [†]	slag	0.0	60.9	34.9	4.3	99.4	1.36	copper	0.00	0.00	100.0	99.6	− 7.50**
		tridymite	0.0	0.69	99.2	0.08	99.7							
9	24 [†]	slag	0.0	61.5	35.2	3.3	100.8	1.36	copper	0.0	0.04	99.9	99.4	− 8.00**
		tridymite	0.0	0.45	99.5	0.07	98.8							
10	24 [†]	slag	0.0	61.3	35.3	3.4	98.9	1.35	copper	0.00	0.00	100.0	99.4	− 8.00**
		tridymite	0.0	0.45	99.5	0.07	98.8							
11	24 [†]	slag	0.0	61.0	36.5	2.5	100.7	1.29	copper	0.0	0.06	99.9	99.7	− 8.50**
		tridymite	0.0	0.84	99.1	0.04	100.6							
12	24 [†]	slag	0.0	61.0	37.4	1.65	100.8	1.27	copper	0.0	0.08	99.9	100.9	− 9.00**
		tridymite	0.0	0.63	99.3	0.04	101.4							
13	24 [†]	slag	0.0	61.1	37.3	1.64	100.7	1.27	copper	0.0	0.09	99.9	101.6	− 9.00**
		tridymite	0.0	0.9	99.1	0.03	100.7							
ZnO-containing system, "Cu ₂ O" in slag = ~ 3.4 wt pct														
14	0.5 [‡]	slag	5.0	69.3	22.4	3.2	99.3	2.68	copper	0.25	0.08	99.6	100.3	− 7.67*
		spinel	2.5	96.8	0.64	0.07	93.1							
15	3	slag	4.3	68.5	24.0	3.1	99.7	2.44	copper	0.22	0.07	99.7	99.2	− 7.73*
		spinel	2.2	97.1	0.56	0.07	92.1							
16	18	slag	16.6	54.8	25.2	3.4	98.9	2.51	copper	0.24	0.09	99.7	100.4	− 7.51*
		spinel	8.0	91.3	0.50	0.06	92.4							
17	2	slag	26.7	45.7	24.3	3.3	100.0	2.83	copper	0.40	0.13	99.5	99.6	− 7.34*
		spinel	12.7	87.1	0.09	0.16	91.2							
		willemite	55.4	15.5	28.9	0.20	101.4							
18	3	slag	16.0	43.0	37.3	3.7	99.7	1.43	copper	0.28	0.36	99.4	100.7	− 7.69*
		tridymite	0.37	0.68	98.6	0.36	99.8							

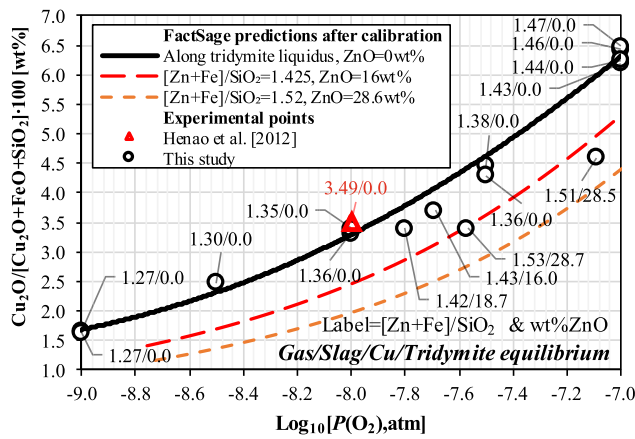
Table IV. Continued

Exp. Code	Time (h)	Phase	Normalized Oxide Composition (Wt Pct)					(Zn + Fe)/ SiO ₂ in Slag	Phase	Normalized Metal Composition (Wt Pct)			Old Total	Log ₁₀ [<i>P</i> (O ₂), Atm]
			ZnO	"FeO"	SiO ₂	"Cu ₂ O"	Old Total			Zn	Fe	Cu		
19	18	slag tridymite	18.7 0.29	39.7 0.56	38.1 98.6	3.4 0.50	99.2 102.3	1.42	copper	0.30	0.05	99.6	99.7	− 7.80*
20	18	slag tridymite willemite	28.7 0.62 57.4	29.6 0.40 13.9	38.3 98.9 28.4	3.4 0.12 0.26	99.4 99.8 101.5	1.53	copper	0.38	0.06	99.6	100.2	− 7.57*
ZnO-containing system, "Cu ₂ O" in slag = ~ 4.4 wt pct														
21	18	slag spinel	18.4 9.1	48.2 90.2	28.3 0.48	5.1 0.14	99.9 92.8	2.13	copper	0.17	0.00	99.8	100.7	− 7.10*
22	18	slag spinel willemite	27.1 14.3 56.2	42.2 85.2 15.2	26.5 0.29 28.2	4.2 0.14 0.37	98.9 92.9 100.7	2.51	copper	0.22	0.44	99.3	100.7	− 7.13*
23	2	slag tridymite willemite	28.5 0.66 57.5	28.6 0.36 13.7	38.2 98.7 28.4	4.6 0.23 0.38	99.2 100.1 100.7	1.51	copper	0.26	0.29	99.4	101.0	− 7.09*
Concentrations of elements below 0.1 wt pct are below the detection limit of the present EPMA measurement.														
*Estimated <i>P</i> (O ₂) based on the calibration curve in Fig. 3 and FactSage prediction using PYROSEARCH database ^[24] (Table III).														
** <i>P</i> (O ₂) was accurately controlled by CO/CO ₂ /Ar gas.														
†Experiments were undertaken using open system equilibration with long equilibration time and accurate <i>P</i> (O ₂) control.														
‡Experiment was undertaken using open system with short time (0.5 h); the test were undertaken using a closed system for longer times.														

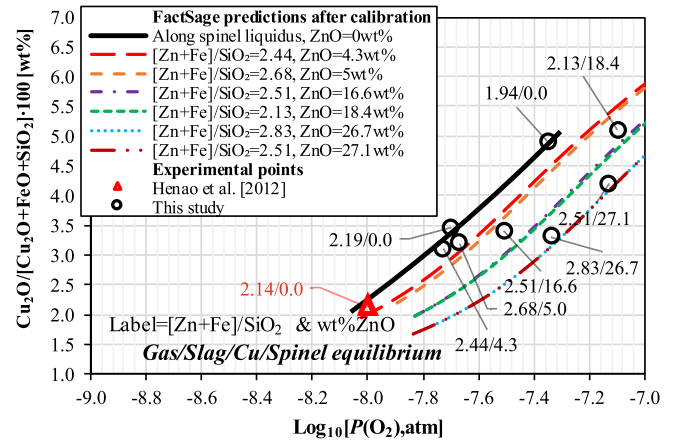
Concentrations of elements below 0.1 wt pct are below the detection limit of the present EPMA measurement.

*Estimated $P(\text{O}_2)$ based on the calibration curve in Fig. 3 and FactSage prediction using PYROSEARCH database^[24] (Table III).** $P(\text{O}_2)$ was accurately controlled by CO/CO₂/Ar gas.†Experiments were undertaken using open system equilibration with long equilibration time and accurate $P(\text{O}_2)$ control.

‡Experiment was undertaken using open system with short time (0.5 h); the rest were undertaken using a closed system for longer times.



(a)



(b)

Fig. 7—“Cu₂O” in slag vs $P(\text{O}_2)$ in the ZnO–“FeO”–SiO₂–“Cu₂O” system at 1250 °C (1523 K): (a) at slag/copper/tridymite equilibrium and (b) at slag/copper/spinel equilibrium. Lines are FactSage predictions with PYROSEARCH database^[24] after calibration. Points are experimental data from the present study and Henao *et al.*^[25] Labels are [Zn + Fe]/SiO₂ ratio and ZnO concentration in slag.

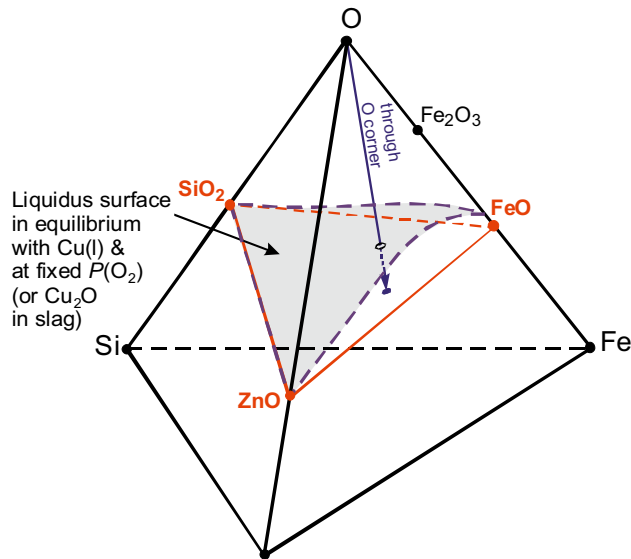


Fig. 8—Schematic position of the liquid oxide in the Zn–Fe–O–SiCu system at fixed “Cu₂O” in slag inside the Zn–Fe–O–Si tetrahedron and projection through the O corner onto the ZnO–“FeO”–SiO₂ plane.

atm) and 4.6 wt pct “Cu₂O” ($P(\text{O}_2) = \sim 10^{-7.1}$ atm) in slag, respectively. At 3.4 wt pct “Cu₂O,” the experimental data obtained from the present study clearly show that the spinel liquidus extends across the pseudo-ternary at almost constant SiO₂ concentration, 0.25SiO₂/(ZnO + “FeO” + SiO₂) by weight. At approximately 0.3ZnO/(ZnO + “FeO” + SiO₂) by weight, the willemite primary phase appears. The willemite liquidus extends from approximately 0.25 to 0.40SiO₂/(ZnO + “FeO” + SiO₂) by weight. The tridymite liquidus is in the range 0.35SiO₂/(ZnO + “FeO” + SiO₂) by weight at 0ZnO to 0.4SiO₂/(ZnO + “FeO” + SiO₂) by weight at 0.30ZnO/(ZnO + “FeO” + SiO₂) by weight. Figure 9

shows that the spinel and willemite phases exhibit solid solutions over wide ranges of compositions. The maximum solubilities of solutes in the spinel and willemite phases (“ZnO” and “Fe₂SiO₄”), respectively, are at the slag/copper/spinel/willemite equilibrium.

IV. DISCUSSION

Figure 10 provides comparison of the liquidus isotherms at 1250 °C (1523 K) for the ZnO–“FeO”–SiO₂ system at $P(\text{O}_2) = 10^{-8}$ atm: (1) predicted using FactSage with the PYROSEARCH database,^[24] (2) reported by Liu *et al.*,^[19] and (3) the experimental liquidus constructed using results from the present study. The ZnO-free liquidus points for the present experimental isotherm were taken from the previous study by the authors.^[26] The ZnO-containing “Cu₂O”-free liquidus at $P(\text{O}_2) = 10^{-8}$ atm was constructed by extrapolation of the present experimental points from the ZnO–“FeO”–SiO₂–“Cu₂O” system using corrections determined by the FactSage predictions with the current thermodynamic database^[24] using the following procedure.

1. Liquidus slag compositions in the ZnO–“FeO”–SiO₂–“Cu₂O” system were calculated initially at the effective $P(\text{O}_2)$'s derived from the “Cu₂O” slag concentrations following the procedure described in Table III (steps *a* through *e* and *k* through *o*).
2. “Cu₂O increments” corresponding to the change of slag from the ZnO–“FeO”–SiO₂–“Cu₂O” to the ZnO–“FeO”–SiO₂ system were predicted for each liquidus slag composition at the derived individual effective $P(\text{O}_2)$'s (Table III, steps *f* through *g* and *p* through *q*).
3. “ $P(\text{O}_2)$ increments,” corresponding to the change of oxidation potential from the individual derived effective $P(\text{O}_2)$ to 10^{-8} atm, in the ZnO–“FeO”–SiO₂ system were predicted for each liquidus slag compo-

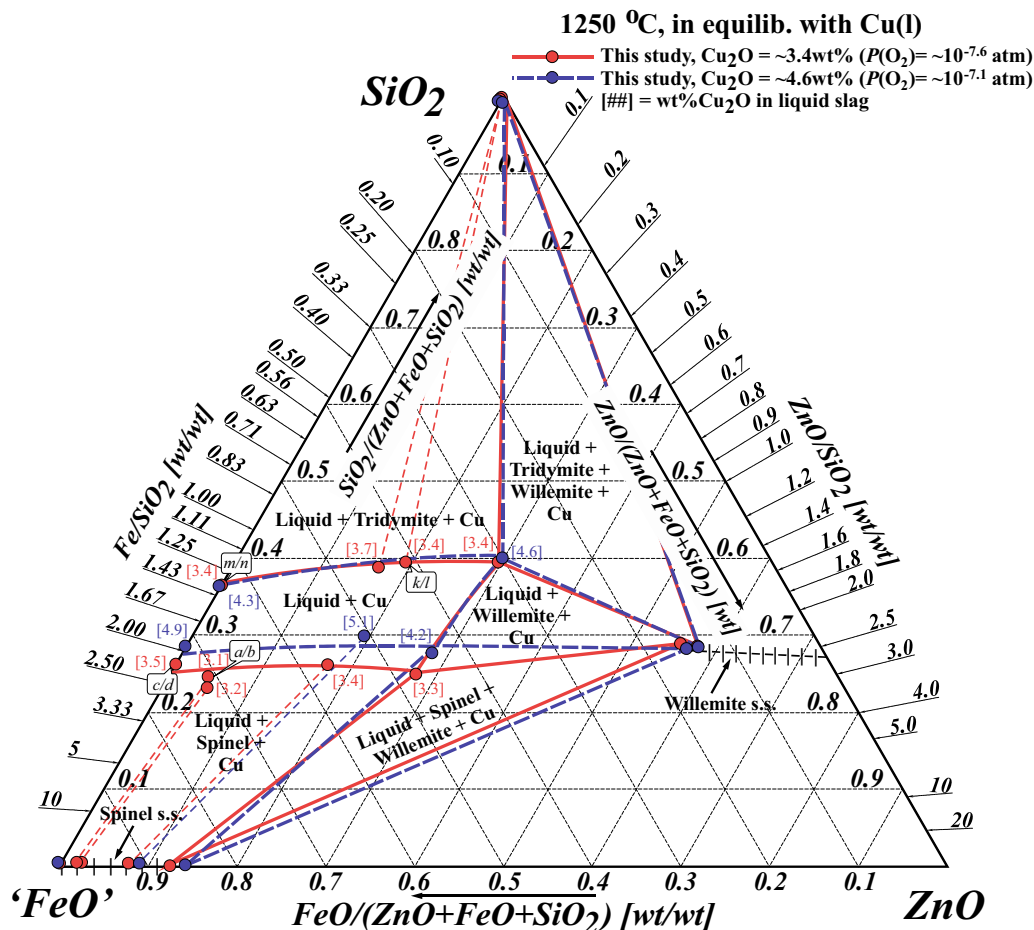


Fig. 9—Phase diagram of the Zn-Fe-O-Si-Cu system in equilibrium with liquid copper at 1250 °C (1523 K) projected onto the ZnO-“FeO”-SiO₂ plane through the oxygen corner. Labels are wt pct Cu₂O in liquid slag.

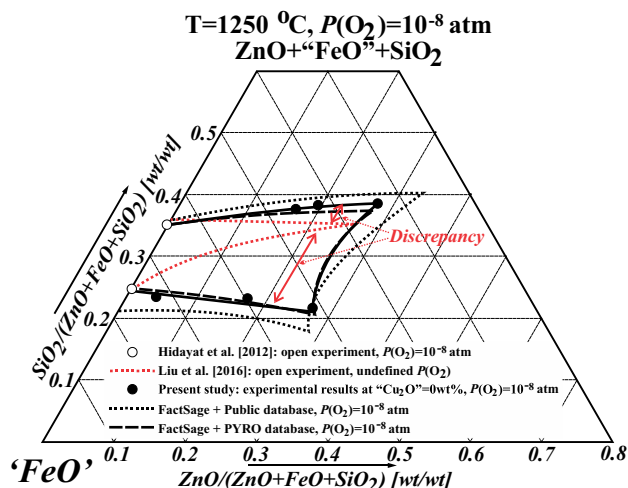


Fig. 10—Liquidus isotherms at 1250 °C (1523 K) for the ZnO-“FeO”-SiO₂ system at $P(\text{O}_2) = 10^{-8}$ atm, calculated with FactSage using public^[11–16] and PYROSEARCH^[24] databases, reported by Liu *et al.*,^[19] and experimentally determined in the present study.

sition (Table III, steps *g* through *h* and *q* through *r*). Each experimental liquidus point from the ZnO-“FeO”-SiO₂-“Cu₂O” system at the derived effective $P(\text{O}_2)$ was then extrapolated to the ZnO-“FeO”-SiO₂ system at $P(\text{O}_2) = 10^{-8}$ atm, applying the individual “Cu₂O” and $P(\text{O}_2)$ increments predicted with FactSage for each point (Table III, steps *i* through *j* and *s* through *t*).

All correction increments were less than 5 wt pct, with the final uncertainty of the experimental liquidus compositions evaluated to be less than 1 wt pct. The validity and low uncertainty of this extrapolation procedure are based on the predictive power of the current FactSage thermodynamic database developed using systematic experimental data on low-dilute “Cu₂O” concentrations in (or close to) the Henrian range and experimental effective $P(\text{O}_2)$'s close to 10^{-8} atm.

The comparison presented in Figure 10 indicates the significant discrepancies between the data reported by Liu *et al.*^[17–21] and the present experimental results. The differences between the liquidus data by Liu *et al.* and the present experimental results increase with the increasing ZnO concentration in the slag. The differences in liquidus compositions of up to 0.1SiO₂/(ZnO + “FeO” + SiO₂) by weight and up to 0.05SiO₂/(ZnO +

“FeO” + SiO₂) by weight are observed in the spinel and tridymite primary phase fields, respectively, and the willemite liquidus was not detected by Liu *et al.* at all. In contrast, the differences between present experimental results and the predictions with the current thermodynamic database^[24] are relatively small.

The present results indicate that the liquidus data reported by Liu *et al.*^[17–21] do not correspond to the reported oxygen pressures and that the effective oxygen activities in the slags in the experiments are unknown. The uncertainties in the results reported by the other authors^[17–21] appear to be the result of uncontrolled Zn (g) vaporization affecting $P(\text{O}_2)$ through Reaction [2]; the observation that the discrepancies increase with increasing ZnO concentration in the slag is consistent with this reaction mechanism.

On this basis, the data reported by Liu *et al.*^[17–21] do not represent the equilibria in these systems for the conditions reported. It is particularly important to obtain accurate data on low-order systems since it is principally from these systems that binary and ternary parameters describing species interactions in complex higher order solutions are derived. In this case, the use of these misleading data for the multicomponent database development could result in the significant distortion of the description of all ZnO-containing slag systems, including significant errors in prediction of slag liquidus temperatures, proportions of solids, and activities of oxygen and of other key components in the slag and condensed phases. All these properties are important to the control and optimization of the actual industrial smelting practices.

The closed system approach used in the present study has considerable potential for extending the range of chemical systems that can be accurately characterized, particularly those systems that contain metal species having relatively high vapor pressures. The technique enables accurate measurement of the distribution of elements between condensed phases in these difficult to characterize systems. In base metal smelting and refining systems, this will include determining the distribution of minor elements between slag/matte/metal/solid phases.

V. CONCLUSIONS

An experimental methodology based on the equilibration in closed system (ampoule), quenching, and EPMA was applied for the Zn-containing slag system having high metal species vapor pressure that provides accurate information on slag phase equilibria at reducing oxygen partial pressures. The technique involves the use of the Cu_{metal}/Cu₂O_{slag} couple as an internal standard to measure the oxygen partial pressure in the slag. Experimental measurements were undertaken to characterize the isothermal section in the ZnO–“FeO”–SiO₂–“Cu₂O”–Cu and ZnO–“FeO”–SiO₂ systems at 1250 °C (1523 K) and $P(\text{O}_2) = 10^{-8}$ atm. The results confirm the accuracy of the thermodynamic model predictions and previous experimental studies^[8–10] undertaken in equilibrium with iron metal and air. The present study demonstrates that the flowing gas

equilibration approach is not suitable for systems in which the metal vapor pressure is high and leads to inaccurate and misleading experimental data.

The study demonstrates the need for identification and critical analysis of elementary reactions, reaction mechanisms, and pathways during equilibration and potential sources of uncertainties before systematic measurements are undertaken, and then for further development of appropriate experimental designs specific to the investigated system to ensure that true chemical equilibrium is achieved. In this way, the generation of misleading and inaccurate data can be avoided and accurate thermodynamic databases developed.

This study is a part of a larger integrated experimental and thermodynamic modeling research program to support the optimization and development of complex lead smelting, refining, and recycling.

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